IPCON® TPU 95A HS

High Speed Polyurethane 95A Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.03 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	320 ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.21 g/cm³
Shrinkage Rate*	Specially-defined	0.18% - 0.37%
Surface Hardness	Shore hardness	95A
Saturated Water Absorption Rate	25 °C, 55% RH, room air	1.04%
Melt Index	230 °C, 2.16 kg	28.4 g/10 min
Melting Temperature	DSC, 10 °C/min	187 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	N / A
Heat Deflection Temperature	ISO 75, 0.45 MPa	N / A

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	27 MPa
Tensile Strength Z	ISO 527, GB/T 1040	22 MPa
Young's Modulus XY	ISO 527, GB/T 1040	16 MPa
Young's Modulus Z	ISO 527, GB/T 1040	13 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	884%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	560%
Impact Strength XY	ISO 178, GB/T 9341	> 125 kJ/m²
Impact Strength Z	ISO 178, GB/T 9341	> 125 kJ/m²
Tearing Strength XY	ISO 34, GB/T 529-2008	112 N/mm
Resilience Rate	ISO 4662, GB/T 1681-2009	30%
Wear Volume (Wear Resistance)	ISO 4649, GB/T 9867-2008	23 mm³
Wear Length	ISO 19252, GB/T 31402-2015	4 mm

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 230 °C, bed temperature: 35 °C, printing speed: XY—144 mm/s and Z—72 mm/s) and IPCON TPU 95A HS (Milky White) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill

pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

Really high speed (up to 300 mm/s), super wear resistance (wear volume is only 23 mm^3), various traditional Chinese colors, easy to print (compatible with most mainstream printers). It can be used to print shoes and bags, mobile phone cases, toy car tires, sealing strips and gaskets, and soft household items.

• Recommended Printing Settings

Drying Settings Before Printing	70 - 90 °C, 8 - 12 h (Blast drying oven)
Printing Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself, PLA or support filament for TPU
Compatible Printer Type	Open-frame, enclosed-frame (Suggest to open the front door and remove the upper lad)
AMS or CFS	Compatible
Compatible Nozzle Material	Any common material, such as brass, stainless steel, hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.2 / 0.4 / 0.6 / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	For most kinds of plate, applying glue is not needed. If the first layer bonding is not firm enough, apply liquid glue or PVP solid glue stick.
Nozzle Temperature	220 - 240 °C
Bed Temperature	30 - 40 °C
Fan Speed	80% - 100%
Max Print Speed	300 mm/s (For the thin and high models and the support structure, the recommended speed is lower than 150 mm/s)

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